

SIMATS ENGINEERING

CO4-ASSESSMENT TOOL2- ARTICLE REVIEW

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1709

COURSE OUTCOME COVERED

CO4: Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)

Assessment Tool Weightage: 10%

Time Duration: 1 Hour

QUESTIONS: 4

Total Marks:40

Code of Conduct:

*I N.Sai Sankar(192311187) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:*N. Sai Sankar*

INSTRUCTIONS

Select any ONE peer-reviewed research article (published after 2019) from IEEE Xplore, Springer, Elsevier, or ACM Digital Library related to any of the following topics: *Inductive Learning, Decision Trees, Statistical Learning Methods (Naïve Bayes / Logistic Regression / SVM), or Reinforcement Learning applications.*

Submit: (i) Article citation details (ii) PDF or valid DOI link (iii) Written review as per the tasks below.

Task 1: Article Summary

(a) Provide the full citation of the article (Author(s), Title, Journal/Conference, Year, DOI). State the domain/application area and the specific ML problem addressed.

(b) Summarise the key objective of the article in your own words (150–200 words). Clearly state the research gap the authors identified and how they proposed to fill it.

Task 2: Methodology Analysis

(a) Explain the ML technique(s) used in the article (e.g., Decision Tree variant, RL algorithm, Statistical model). Describe the dataset used—size, features, source, and how it was prepared.

(b) Map the technique to CO 4 topics (Inductive Learning / Decision Trees / Statistical Learning / RL). Identify one methodological strength and one limitation in the authors' approach.

Task 3: Results & Critical Evaluation

(a) Report the key results (accuracy, F1-Score, AUC-ROC, reward convergence, etc.) as presented by the authors. Create a comparative table if multiple models are benchmarked.

- (b)** Critically evaluate the results: Are the reported metrics realistic? Are there potential biases (dataset, evaluation protocol)? What additional experiments would strengthen the findings?
- (c)** Discuss real-world applicability: Where can this technique be deployed? What are the challenges in scaling the solution to industry (data availability, compute, ethical issues)?

Task 4: Reflection & Learning Outcome

- (a)** State three key insights you gained from reading the article that deepen your understanding of CO 4 topics. Connect each insight to a specific concept from the course syllabus.
- (b)** If you were to extend this research, propose one novel experiment or enhancement (different dataset / improved algorithm / new application domain). Justify its feasibility and expected impact.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article's purpose, concepts, and arguments	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand

1(a) Citation, Domain and ML Problem

Full Citation:

Almustafa, K. M. (2020). *Prediction of heart disease and classifiers' sensitivity analysis*. BMC Bioinformatics, 21, 278. <https://doi.org/10.1186/s12859-020-03626-y>.

Domain/Application:

Healthcare and medical diagnosis.

Machine Learning Problem:

The research focuses on using machine learning classification algorithms to predict whether a patient has heart disease. The study particularly investigates whether heart disease can be predicted accurately using a smaller number of medical attributes.

The authors compare K-NN, Naïve Bayes, Decision Tree J48, JRip, SVM, AdaBoost, SGD and Decision Table classifiers.

1(b) Article Summary – 150–200 words

The article “Prediction of heart disease and classifiers’ sensitivity analysis” investigates the use of machine learning classification techniques for predicting heart disease. The main objective is to determine which classification algorithm can accurately classify heart-disease cases while using a minimum number of medical attributes. The authors use a heart-disease dataset containing 1025 patient records. Although the original dataset contains 76 attributes including the class attribute, the study uses 14 selected attributes, including age, sex, chest pain type, blood pressure, cholesterol, fasting blood sugar, ECG results, maximum heart rate, exercise-induced angina, oldpeak, slope, number of major vessels, thalassemia and the target class.

The researchers compare several machine-learning classifiers, including K-NN, Naïve Bayes, Decision Tree J48, JRip, SVM, AdaBoost, SGD and Decision Table. Ten-fold cross-validation is used for performance evaluation. The research gap addressed is the need to identify effective classifiers and a smaller set of relevant attributes for heart-disease prediction. The authors also perform sensitivity analysis on the J48 pruning confidence factor and apply feature selection. The results show that J48 achieved 98.0488% accuracy, while optimized pruning improved it slightly to 98.1463%.

2(a) ML Techniques Used

The article uses multiple supervised classification techniques:

1. K-Nearest Neighbor (K-NN)
2. Naïve Bayes
3. Decision Tree J48
4. JRip
5. Support Vector Machine (SVM)
6. AdaBoost
7. Stochastic Gradient Descent (SGD)
8. Decision Table

The primary technique relevant to CO4 is Decision Tree J48. J48 is an implementation of the C4.5 decision-tree algorithm. It recursively divides the dataset using decision conditions until classification decisions can be made.

Dataset

The dataset contains:

- 1025 patient records
- Original dataset: 76 attributes including the class
- Attributes used in the study: 14
- Prediction class: presence/absence of heart disease
- Data sources: Cleveland, Hungary, Switzerland and Long Beach datasets.

Main Features

No.	Feature
1	Age
2	Sex
3	Chest pain type
4	Resting blood pressure
5	Serum cholesterol
6	Fasting blood sugar
7	Resting ECG results
8	Maximum heart rate
9	Exercise-induced angina
10	Oldpeak
11	Slope

12	Number of major vessels
13	Thalassemia/defect
14	Class/target

The researchers used WEKA 3.8.4 and evaluated the classifiers using 10-fold cross-validation.

Feature Selection

A Classifier Subset Evaluator was used to identify relevant features. The study found that some classifiers could obtain very high performance using only a small subset of attributes. For example, the selected important attributes included age, chest pain type, cholesterol and maximum heart rate.

2(b) Mapping to CO4

CO4 Topic	Connection with Article
Inductive Learning	The classifiers learn patterns from existing patient examples and use them to predict unseen cases.
Decision Trees	J48 decision tree is directly implemented and evaluated.
Statistical Learning	Kappa, ROC, MAE and RAE are used to evaluate classification performance.
Classification	The objective is to classify patients into heart-disease and non-heart-disease classes.
Feature Selection	Relevant attributes are selected to reduce the number of features required for prediction.

3(a) Key Results

The main classifier results reported in the article include:

Classifier	Accuracy	Kappa	ROC
K-NN (K=1)	99.7073%	0.9941	0.994
Decision Tree J48	98.0488%	0.961	0.996
JRip	97.2683%	0.9454	0.996
Other classifiers	Lower than the above	—	—

The reported confusion matrix for J48 is:

	Predicted 0	Predicted 1
Actual 0	497	2
Actual 1	18	508

This gives an accuracy of 98.0488%.

J48 Sensitivity Analysis

The researchers varied the Pruning Confidence Factor (PCF).

PCF	Accuracy
0.25	98.0488%
0.30	98.1463%
0.35	98.1463%

Thus, changing the pruning parameter from 0.25 to 0.30/0.35 produced a small improvement in J48 accuracy.

Feature Selection Results

Feature selection produced interesting results. K-NN increased from 99.7073% to 100%, while J48 changed from 98.0488% to 97.6585% after feature selection.

3(b) Critical Evaluation

Are the Metrics Realistic?

The reported accuracy values are very high, particularly the 99.7073% K-NN accuracy and the 98.0488% J48 accuracy. These results are promising, but they should not automatically be interpreted as clinical-level performance.

Accuracy alone may not be sufficient for medical diagnosis. Metrics such as sensitivity, specificity, precision, F1-score and ROC-AUC should also be examined.

Potential Biases

Possible limitations include:

1. Dataset size: 1025 records may be insufficient for broad clinical generalization.
2. Dataset composition: Data originate from specific medical centers.
3. Feature limitations: Only 14 attributes were used from the original 76.
4. Cross-validation: Ten-fold cross-validation is useful but does not replace external validation.
5. Possible overfitting: Very high accuracy can indicate that the model performs particularly well on the available dataset.

3(c) Real-World Applicability

The technique could potentially be deployed as a clinical decision-support system. A doctor could enter basic patient information such as age, chest-pain type, cholesterol level and maximum heart rate, and the system could provide a preliminary risk classification.

Possible Applications

- Hospital decision-support systems
- Preliminary heart-disease screening
- Telemedicine systems
- Mobile healthcare applications
- Rural healthcare centers
- Patient risk-monitoring systems

Industry Challenges

1. Data availability:

Large, high-quality and representative medical datasets are difficult to obtain.

2. Privacy:

Patient medical information must be protected.

3. Generalization:

A model trained on one population may not perform equally well on another population.

4. Explainability:

Medical professionals need understandable reasons for predictions.

TASK 4 – REFLECTION & LEARNING OUTCOME

4(a) Three Key Insights

Insight 1 – Decision Tree Classification

I learned that Decision Trees can classify medical data by repeatedly dividing the dataset according to important attributes. The article's use of J48, a C4.5-based decision tree, helped me understand how decision-tree classification can be applied to real-world problems.

CO4 Connection: Decision Tree Algorithms.

Insight 2 – Feature Selection

The article demonstrated that using fewer features can sometimes maintain or improve classification performance. The authors identified combinations such as age, chest pain type, cholesterol and maximum heart rate as highly useful features.

CO4 Connection: Inductive Learning and Feature Selection.

Insight 3 – Model Evaluation

I learned that accuracy alone cannot completely describe the quality of a machine-learning classifier. The researchers also considered Kappa, ROC, MAE and RAE to evaluate the models.

CO4 Connection: Statistical Learning and Classification Evaluation.

4(b) Proposed Novel Experiment

Proposed Experiment: External Validation with a Multi-Hospital Dataset

I would extend this research by testing the J48 model on a larger and independent multi-hospital heart-disease dataset.

The experiment would use the same selected features—age, chest pain type, cholesterol and maximum heart rate—and compare J48 with Random Forest, XGBoost, SVM and K-NN. Stratified 10-fold cross-validation would be performed on the training dataset, followed by external testing on data from another hospital.

This enhancement is feasible because the original study already demonstrates that a small number of attributes can provide strong predictive performance. External validation would determine whether the reported performance generalizes beyond the original dataset. It would also provide stronger evidence for possible healthcare deployment.